

[DC] Improving Guidance for Pedagogical Agents in Educational VR

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ABSTRACT

Virtual Reality (VR) can create engaging and interactive learner experiences, but very few studies explore the importance of agent guidance considerate to the learner's attention in educational VR. In my dissertation, I investigate how to use combined motion and attention-based physiological responses to improve pedagogical agent response and guidance in VR and enhance educational VR experiences. This work's main novelty is the creation of the first Responsive Pedagogical Agent for Educational VR (RePAEVR) system which provides insights into how to design, measure, and communicate pedagogical agent guidance in educational VR.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools

1 INTRODUCTION

Imagine having a virtual assistant in an educational VR environment that understood what could help you learn better, how to keep you from losing interest, and that could adapt their instruction just like an expert human instructor could. My dissertation explores creating personalized agent-guided educational VR experiences.

Virtual learning environments have used responsive systems such as intelligent tutoring systems (ITS) and pedagogical agents. Such approaches rely on adapting instruction by providing interactive feedback to guide learners based on a student model or student errors. These improve learner perceptions; however, results on learning continue to be inconclusive [5]. Student errors (e.g., test scores) are useful for the overall experience but attention and learning can vary over time. 2D interfaces limit interactivity of agent guidance, requiring external sensors for more elaborate behaviors.

Past researchers measure attention by using eye-tracking metrics and provide a minimal response if attention shifts, such as a warning [3]. Performance metrics such as test scores assess information retention, but this does not account for situations such as a learner paying attention but struggling to comprehend bits of critical knowledge. In this case immediate and personalized agent feedback, such as providing contextual guidance or hints, can be more appropriate for addressing misunderstandings and preventing disengagement.

The main contribution of my work is the design, development, evaluation, and demonstration of a responsive pedagogical agent for an educational VR (RePAEVR) system that adapts agent guidance in real time based on learner attention and learning. Benefits from my research can advance immersive 360° or VR environments in multiple domains such as tourism, gaming, and training.

My main research question is: "How do we adapt a pedagogical agent's guidance in VR to the learner's attention and performance in

real time?" This question is broken down into three sub-questions: (1) "How do we measure physiological responses related to attentiveness in VR?" (2) "How do we measure biometric data related to a person's learning performance in VR?" and (3) "How do we design and implement a responsive pedagogical agent architecture for educational VR?"

2 BACKGROUND

Virtual Reality (VR) opens up new opportunities with interactive and immersive learning experiences. Affordable consumer-level VR headsets with varied sensing (head, eye, facial, and hand tracking) such as the Meta Quest Pro, allow for varied sensor and motion data to be readily available in interactive VR environments.

There has been a lot of research investigating techniques to improve pedagogical agents to express emotions or provide nonverbal cues through body language, such as gestures or head nods [5]. Less work has been done to adapt instruction to physiological responses. D'Mello et al. [2] designed an ITS with gaze-reactive guidance to reorient a user's attention in a 2D desktop interface.

Physiological sensing can extend attention detection, such as electrodermal activity (EDA) or electroencephalography (EEG). A multimodal deep fusion approach to predicting learner attention using physiological signals can combine and classify data from multiple biometric sensors. One study integrated multimodal deep fusion to detect cybersickness in VR [4] with a CNN-LSTM and 3D-CNN ensemble. Therefore we aim to address this gap in attention detection by integrating and measuring ECG, EEG, head, eye, and hand tracking data in a multimodal deep learning approach.

Researchers have explored predicting learning in VR by classifying learners based on motion. Amershi et al. [1] predicted low or high learning with an accuracy of 86.3% by clustering on gaze motion using k-means. Moore et al. [8] predicted learning levels in a VR training environment using an SVM classifier on motion data with an accuracy of 86.7%, finding that slower and abrupt movements were associated with low learning gains. Motion plays a crucial role in learning, yet minimal research has been done to improve agent behaviors based on predicting learning from motion data. We aim to address this gap with realtime learning prediction using motion data.

3 WORK TO DATE

To get familiar with pedagogical agent design and deep learning classifiers, I conducted two VR studies (Figure 2). The first study designed RePAEVR's architecture (Figure 1) and used it to improve pedagogical agent spatial guidance behaviors in educational VR. The second study extended our architecture's sensing with a CNN-LSTM classifier to detect distraction.

In the first study [6], I improved agent response with interactive gaze-reactive behaviors during spatial guidance. Attentional shifts are sensed through generalized hotspots in our architecture that combine low-level sensor information into distraction scores. Gaze-reactive agent behavior is sequenced by processing a response description containing candidate responses and associated priority ranks modified by distraction scores computed from 3D hotspots. I looked at some teacher agent behaviors and found that gaze-reactive

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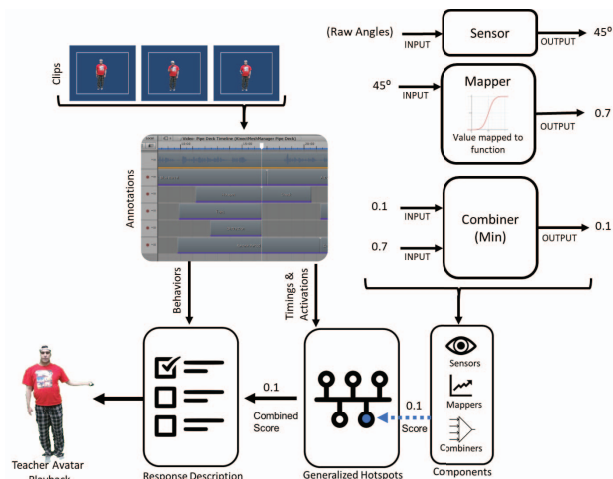


Figure 1: Architecture for educational VR pedagogical agents.

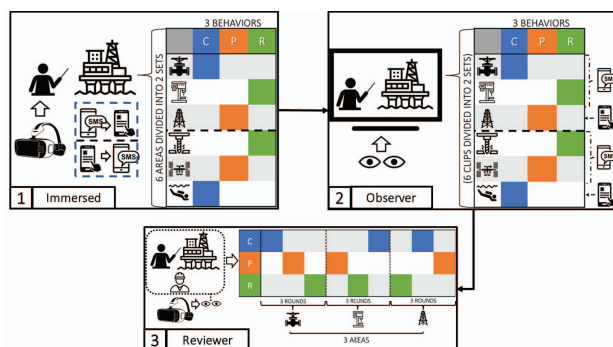


Figure 2: Experiment diagram for both studies.

agent guidance considering more interactivity was preferred. This investigated RQ3.

Next, I explored if distraction could be predicted from generalized motion features (angular and positional velocities) extracted from VR headset head-tracking, hand-tracking, and eye-tracking sensor data. I used a deep learning approach using a CNN-LSTM neural network to train a classifier, achieving an accuracy of 94.93% [7]. This investigated RQ1.

3.1 Future Work

I will investigate using motion data to predict learning performance in educational chemistry VR app (RQ2). In the app, molecules react in specific configurations (relative angles, distances) between atoms to mimic real-world conditions. By manipulating molecular configurations with hand movements, students can experiment with configurations and receive interactive feedback on their understanding of reactions.

Data collected will be analyzed for features that predict success at identifying reactions. I will train a classifier to predict learning performance based on this data. For RQ1, I will extend my CNN-LSTM classifier with a multimodal architecture to detect inattention by incorporating physiological signals (EEG, ECG, HRV).

Finally, I will develop RePAEVR real-time guidance based on results from RQ1 and RQ2 by extending the RePAEVR architecture from the first study so 3D hotspot distraction scores take into account learning performance levels and multimodal deep learning

attention prediction. This will provide real-time agent guidance. I will be testing the hypothesis that RePAEVR will improve learning by providing a more engaging and immersive experience.

4 EXPECTED CONTRIBUTIONS

My Ph.D. research contributions are in two primary categories.

4.1 Systems Contribution

(1) Pedagogical agent architecture for educational VR environments that increases interactivity of agent guidance behaviors based on 3D hotspots and distraction scores. (2) A multimodal system using varied VR headset and physiological sensing for capturing attention and learning in educational VR. (3) The RePAEVR system that can adapt instruction in realtime and provide personalized guidance, such as hints or simple examples, based on attention and learning.

4.2 Empirical Contribution

Beyond contributing implementations, (1) Experiment design method in VR to evaluate gaze-reactive agent guidance behaviors. (2) Methods for extracting a user's contextual information, such as attention and learning, from motion and physiological data to provide context-aware recognition of learner state. (3) Findings from user studies conducted using the RePAEVR system to contribute to knowledge about context-aware pedagogical agent guidance in educational VR.

5 EXPECTED BENEFITS OF PARTICIPATION

Advice from experts in the IEEEVR community DC consortium will greatly strengthen my dissertation and help me form connections for potential collaborations. I will contribute to discussions with fellow consortium participants to provide unique perspectives for others to develop and improve their research. I am looking forward to the opportunity to work with the DC consortium and its members.

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